Experiment-1

Name: -P.Divyashree **Regno: -**192472291

Course Code: -MLA0305 **Slot-**B

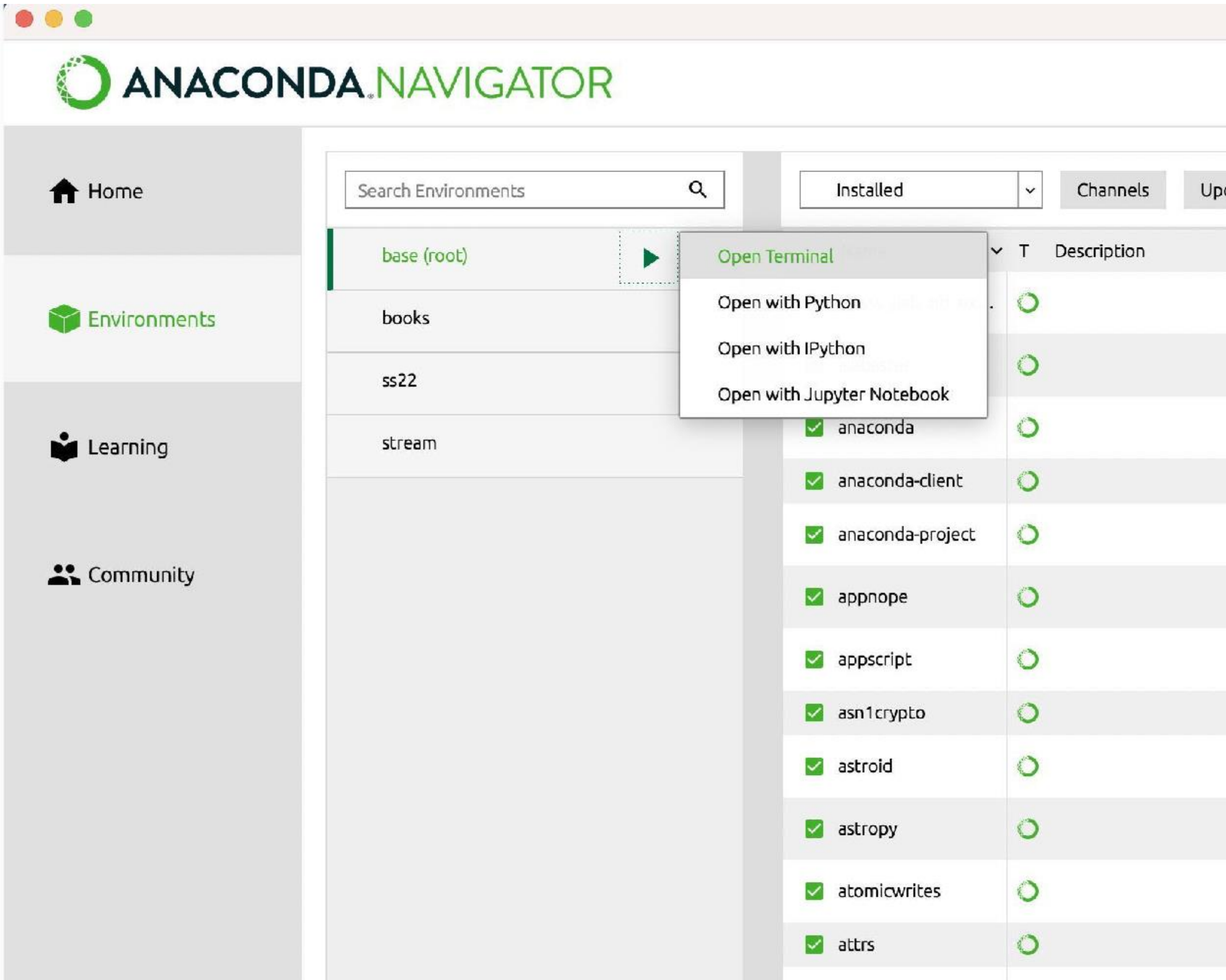
Title: -Installation and Configuration of Anaconda Navigator, Python, TensorFlow, Keras and Gymnasium

AIM

To install and configure Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium, and to verify that all the required tools and libraries are working correctly for Reinforcement Learning experiments.

PROCEDURE

1. Install Anaconda Navigator in the system.
2. Open Anaconda Prompt and verify the Conda installation using `conda--version`.
3. Check the installed Python version using `python --version`.
4. Create a separate environment for Reinforcement Learning using:
   `conda create -n rein python=3.11 -y`
5. Activate the environment using:
   `conda activate rl_env`
6. Install the required libraries such as NumPy, Pandas, Matplotlib and Scikit-learn.
7. Install TensorFlow using `pip install tensorflow`.
8. Install Keras using `pip install keras`.
9. Install Gymnasium using `pip install gymnasium`.
10. Open Jupyter Notebook through Anaconda Navigator.
11. Execute the verification program.
12. Verify the versions of Python, NumPy, TensorFlow, Keras and Gymnasium.
13. Perform a TensorFlow matrix multiplication test.
14. Create the `CartPole-v1` environment using Gymnasium.
15. Verify the observation space, action space, sample action and reward.
16. If all tests execute successfully, the Reinforcement Learning environment is considered properly configured.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Description
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anaconda-client
anaconda-project
appnope
appscript
asn1crypto
astroid
astropy
atomicwrites
attrs

Launcher | Untitled1.ipynb

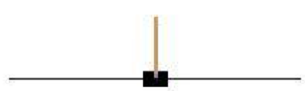
Code git

[2]:

```python
import tensorflow as tf

# Check if TensorFlow can access the GPU
print("Num GPUs Available: ", len(tf.config.list_physical_devices('GPU')))
print("Available GPUs: ", tf.config.list_physical_devices('GPU'))

# Basic TensorFlow computation to utilize GPU
# Create two tensors
a = tf.constant([[1.0, 2.0], [3.0, 4.0]])
b = tf.constant([[1.0, 1.0], [0.0, 1.0]])

# Run a matrix multiplication (GPU should handle this if available)
c = tf.matmul(a, b)

print("Result of matrix multiplication:\n", c)
```

```
Num GPUs Available:  1
Available GPUs:  [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
Result of matrix multiplication:
 tf.Tensor(
[[1. 3.]
 [3. 7.]], shape=(2, 2), dtype=float32)
```

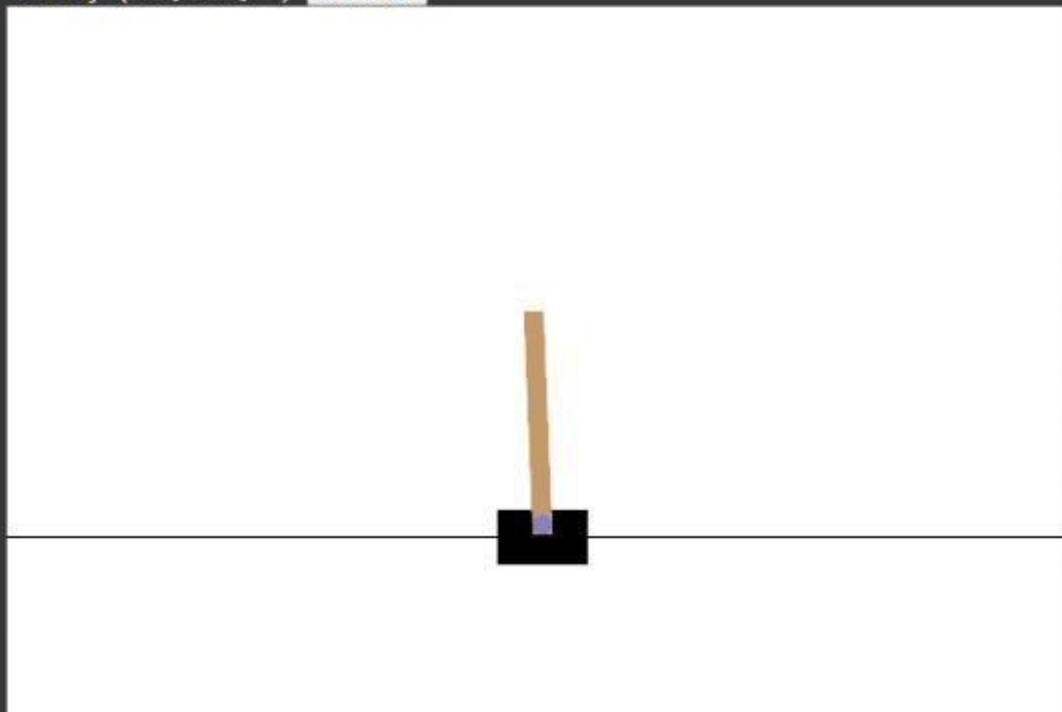
Cart Pole



This environment is part of the Classic Control environments which contains general information about the environment.

Action Space	`Discrete(2)`
Observation Space	`Box([-4.8000002e+00 -3.4028235e+38 -4.1887903e-01 -3.4028235e+38], [4.8000002e+00 3.4028235e+38 4.1887903e-01 3.4028235e+38], (4,), float32)`
import	`gymnasium.make("CartPole-v1")`

```
(array([ 0.01730201, -0.02090577, -0.03592347,  0.00567385], dtype=float32), {})
ndarray (400, 600, 3) show data
```



RESULT

The required Reinforcement Learning environment was successfully configured.